WEEK 2

Cracks in Classical Physics

25, 27 August · Objectives 1, 2

Three experiments broke the continuum: blackbody radiation, the photoelectric effect, and double-slit interference with single particles. The first two force energy quantization. The third forces something harder — that a single indivisible object carries a distribution over paths, and that *learning which path* destroys it.

Blackbody radiation and the ultraviolet catastrophe

A blackbody absorbs all incident radiation and re-emits based only on temperature. Classical statistical mechanics assigns each electromagnetic mode in a cavity an average energy of kT (equipartition). The mode count grows as ν^2, so the predicted spectral density is:

```
Rayleigh-Jeans:   ρ(ν) = (8πν²/c³) · kT        →  ∫ρ dν = ∞
```

The integral diverges. A warm cavity would radiate infinite energy, concentrated at short wavelengths. Ehrenfest named it the ultraviolet catastrophe in 1911. It is not a small discrepancy — it is a theory returning infinity where an oven returns a dull red glow.

Planck's 1900 fix: permit mode energies only in integer multiples of $h\nu$.

```
Planck:   ρ(ν) = (8πν²/c³) · hν/(e^{hν/kT} - 1)
```

The exponential suppresses high-frequency modes. When $h\nu \gg kT$, thermal energy cannot pay the entry cost of even one quantum, and the mode stays silent. This is a **minimum transaction size**, and it is the entire quantum hypothesis.

Planck regarded it as a mathematical trick for six years and disliked it. The constant that fell out — $h = 6.626 \times 10^{-34}$ J·s — is now the definition of the kilogram.

> **Analogy boundary — quantization as discretization.** Energy quantization resembles fixed-point arithmetic: values on a lattice, not a continuum. It fails on two counts. First, the spacing hν is not universal — it scales with frequency, so different modes have different grid sizes. Second, and more importantly, a quantum system can occupy a **superposition** of lattice points, which no fixed-point register can. Quantization constrains measurement *outcomes*, not the state.

The photoelectric effect

Light on metal ejects electrons. Classically, energy arrives continuously, so brighter light should eject faster electrons, and dim light should work after a delay while energy accumulates.

Observed instead:

```
 OBSERVATION                           CLASSICAL EXPECTATION
─────────────────────────────────────────────────────────────────
Below threshold ν₀: nothing,          Any frequency works given
  at any intensity                      enough intensity
Above ν₀: KE depends on ν only        KE rises with intensity
Intensity sets electron COUNT         Intensity sets electron ENERGY
Emission is immediate (< 10⁻⁹ s)      Dim light needs seconds to build
```

Einstein's 1905 account — the work that earned his Nobel, not relativity — treats light as quanta of energy hν:

```
KE_max = hν - φ          φ = work function of the metal
```

The logic is per-photon. One photon either clears the work function or does not. A brighter beam of insufficient photons is more failed attempts, not one success.

> **Analogy boundary — photons as packets.** The packet framing gets threshold behavior exactly right: a message under the minimum size is rejected regardless of send rate. It fails on identity and locality. Photons are excitations of a field mode, not localized objects in flight — they have no trajectory between emission and absorption, and they are strictly indistinguishable in a way no packet with a sequence number is. Push the packet analogy into the double-slit and it produces the wrong answer, as the next section shows.

The double-slit experiment

Young established the wave nature of light in 1801 by passing it through two slits and observing fringes. Davisson and Germer accidentally did the same for electrons in 1927, confirming de Broglie's 1924 hypothesis that matter has wavelength $\lambda = h/p$.

The version that matters fires particles **one at a time**:

```
      SOURCE        BARRIER         SCREEN
                   ┌──────┐        ░░▓▓░░▓▓░░
         ·     ──► │  ══  │   ──►  ░▓▓██▓▓██▓░
     (one          │      │        ░░▓▓░░▓▓░░
      electron)    │  ══  │        ▒▒░░▒▒░░▒▒
                   └──────┘
                                   fringes emerge over
                                   thousands of single hits
```

Each electron makes one localized dot. The dots accumulate into interference fringes. The electron is indivisible — it never splits — yet the pattern requires contributions from both slits. A single electron's amplitude passes through both.

Now the part that makes this a quantum-*information* result. Place a detector at one slit to record the path. The fringes vanish, replaced by two classical bands. And this holds **regardless of detector mechanism** — the disturbance can be made arbitrarily gentle. What destroys interference is not momentum kick but the *existence of which-path information* anywhere in the universe.

Formally: interference requires the two paths to be **indistinguishable**. If path information is imprinted on any system, the branches no longer share a final state and cannot interfere:

```
   Interference:      (|L⟩ + |R⟩)/√2                  → fringes
  Path recorded:     (|L⟩|d_L⟩ + |R⟩|d_R⟩)/√2         → no fringes,
                                                         because ⟨d_L|d_R⟩ = 0
```

The fringes return if you erase the detector record before it becomes permanent — the quantum eraser. Interference tracks *information availability*, not physical roughness.

> **Analogy boundary — which-path as an audit record.** This is the strongest analogy in the course and worth stating precisely. Writing a which-path record is exactly like appending to an audit trail: it makes one branch of a formerly-open computation permanently distinguishable and forecloses the alternative. The boundary: your audit entry records a fact that was already true. The which-path detector does not record a pre-existing path — before entanglement with the detector, the electron had no path. The record **creates** the fact it appears to log. That has no classical counterpart, and it is why "hidden variables" is a substantive claim rather than a truism.

> **Try it yourself.** Open `double_slit.html`. Run in particle mode and watch the fringes build from individual hits — note that the first hundred look random. Then enable the which-path detector and watch the pattern collapse to two bands. The transition is not gradual in detector strength; it tracks how much information the detector actually acquires.

Bohr's atom

Rutherford's 1911 nuclear atom is classically unstable: an orbiting electron accelerates, radiates, and spirals in within ~10^{-11} seconds. Matter should not exist.

Bohr's 1913 postulate: angular momentum is quantized, $L = n\hbar$. Electrons occupy stationary states and radiate only when transitioning between them.

```
E_n = -13.6 eV / n²          ΔE = E_final - E_initial = hν
```

This predicts hydrogen's spectral lines to four significant figures — the Balmer series falls out exactly. It is also, as Bohr knew, internally incoherent: it grafts a quantization rule onto classical orbits without justifying either. It fails for helium.

De Broglie supplied the missing why in 1924. If electrons have wavelength $\lambda = h/p$, a stationary orbit is one holding a standing wave — an integer number of wavelengths around the circumference:

```
nλ = 2πr    with λ = h/p    ⟹    L = pr = nħ
```

Bohr's postulate is a boundary condition on a wave. This is the last step before Schrödinger replaces orbits with orbitals in Week 3.

Key takeaways

- Blackbody radiation forces a minimum energy transaction of hν per mode; the exponential suppression of costly modes resolves the divergence.
- The photoelectric effect is per-photon logic: threshold frequency, intensity setting count rather than energy, and instant emission.
- Single-particle double-slit interference shows one indivisible object carrying amplitude over multiple paths.
- Which-path information destroys interference independent of mechanism. Availability of information, not disturbance, is the operative variable.
- Bohr's quantized orbits reproduce hydrogen's spectrum; de Broglie explains them as standing waves, converting an ad hoc rule into a boundary condition.

Self-check

Q1 *(Objective 2)* — A quantum eraser recovers interference by destroying which-path information *after* the particle has already hit the screen. Your instinct as an audit-systems engineer is that this is retrocausal — the past being rewritten. Explain why it is not, in terms of what is and is not recoverable from the raw hit data.

Q2 *(Objectives 1, 2)* — Bohr's model predicts hydrogen's spectrum to four significant figures and fails for helium. In your own architectural vocabulary, characterize what kind of model this is, and say why its numerical success made it more dangerous than an obviously wrong model.

Check your answers

A1 — Nothing about the past changes, and the resolution lies in **what the screen data alone contains**.

The full set of hits shows no interference — it is a smooth blob, both when path information is kept and when it is later erased. Fringes appear only in **subsets** of the data, selected by the detector's outcome. Erasing which-path information does not restore fringes to the existing pattern; it makes available a *sorting key* that partitions the hits into two subensembles, each of which shows fringes, with the two sets phase-shifted so their sum is the featureless blob you already had.

The audit framing that makes this click: you are not rewriting log entries. You are gaining the ability to `GROUP BY` a column that was always in the data. The correlations existed at hit time; what the erasure changes is your access to the key. This is exactly why no signal can be sent — the marginal distribution at the screen is fixed the instant the particle lands, and a distant experimenter's choice cannot alter it. Retrocausality would require the marginal to change; it does not.

A2 — It is a **model fit to one instance rather than derived from an interface** — a lookup table with a plausible-looking closed form. Bohr had a correct constraint (quantization) attached to an incorrect mechanism (classical orbits), tuned against the one system whose spectrum he had: hydrogen, the single-electron case where electron–electron interaction is absent and the error is invisible.

The four-significant-figure agreement is the danger. A model that is obviously wrong is discarded cheaply. A model that is precisely right in the tested regime and structurally wrong outside it acquires institutional weight — it gets taught, extended, and defended. The failure at helium was the load test that revealed the fit had no mechanism behind it.

The recognizable pattern: a component passing every integration test because the suite covers one code path. Bohr, to his credit, said so himself — he called the model provisional from the start and spent a decade seeking the theory beneath it. The proper fix was not patching the orbits but replacing the abstraction, which is what Schrödinger and Heisenberg did in 1925–26.